

Are galaxies in compact groups special?

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ABSTRACT

We investigate the properties of galaxies in compact groups (CGs) and compare them to a control sample of galaxies.

Key words. galaxies: clusters: general – catalogs

1. Introduction

2. Data

2.1. Samples

We use the compact groups and control samples we built in our previous article [Tricottet et al. \(2025\)](#). We nevertheless briefly recall here how they were built. [Explain.](#) Contrarily to our previous article, though, we remove CGs that we classified as *split* following [Zheng & Shen \(2021\)](#). In this process, our initial compact group sample of 78 groups reduced to 62. We also remove groups bluit from Lim group 3688 from our Control_{4B} and Control_{4C} samples. This group has six galaxies having a redshift between 0.032 and 0.034, and one having a redshift of 0.048. This group seems spurious, and the outlier galaxy, although not being the BGG, is selected both in our original Control_{4B} and Control_{4C}, influencing the velocity dispersion of the group significantly (1979 and 1981 km s⁻¹, respectively).

We use the SDSS DR 16 to retrieve masses, star formation rates (SFRs) and morphologies of galaxies assessed in Galaxy Zoo [Lintott et al. \(2011\)](#). We specifically extracted fields `sfr_tot_p50`, `specsfr_tot_p50` and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table.

2.2. sSFR

To build a general star formation classification that could be uniformly applied to all our samples, we extracted galaxies from the SDSS DR 16 through the query displayed in listing 1.

```
p.petroMag_r,petroMag_u,petroMag_g,
petroMag_i,petroMag_z,
p.objID,
g.sfr_tot_p50, g.specsfr_tot_p50, g.
lgm_tot_p50,
l.h_alpha_eqw, l.h_beta_eqw, l.
oiii_5007_eqw, l.nii_6584_eqw,
l.h_alpha_flux, l.h_beta_flux, l.
oiii_5007_flux, l.nii_6584_flux,
z.p_el_debiased AS p_E,
z.p_cs_debiased AS p_S
FROM SpecObj AS s
JOIN PhotoObj AS p ON s.bestObjID = p
.objID
JOIN galSpecExtra as g ON s.specObjID
= g.specObjID
JOIN galSpecLine as l ON s.specObjID
= l.specObjID
JOIN zooSpec AS z ON s.specObjID = z.
specObjID
WHERE s.z BETWEEN 0.005 AND 0.0452
AND (p.petroMag_r - p.extinction_r <=
17.77)
AND s.class = 'GALAXY'
AND g.lgm_tot_p50 > -1000
```

Listing 1: Query used for selecting spectral & photometric data from the SDSS

We select non-AGN galaxies by first placing them on the classical BPT diagnostic of [Veilleux & Osterbrock \(1987\)](#) and requiring measured values of both emission-line ratios

$\log_{10}([\text{N II}]\lambda 6584/\text{H}\alpha)$ and $\log_{10}([\text{O III}]\lambda 5007/\text{H}\beta)$.

A galaxy is flagged as an AGN if it satisfies either

$$\log_{10} \frac{[\text{N II}]}{\text{H}\alpha} > 0,$$

or if it lies above the empirical demarcation of [Kauffmann et al. \(2003\)](#):

$$\log_{10} \frac{[\text{O III}]}{\text{H}\beta} > \frac{0.61}{\log_{10}([\text{N II}]/\text{H}\alpha) - 0.05} + 1.3,$$

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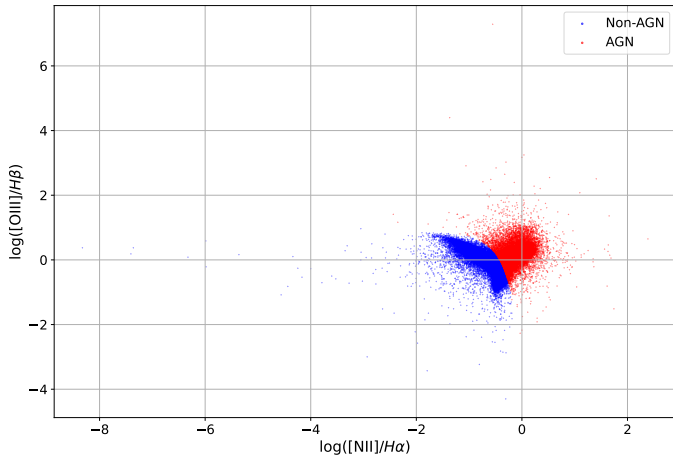


Fig. 1: BPT diagram showing our separation between ordinary and AGN galaxies

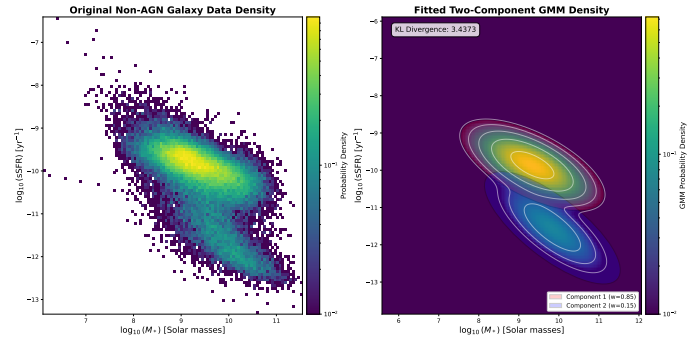


Fig. 2

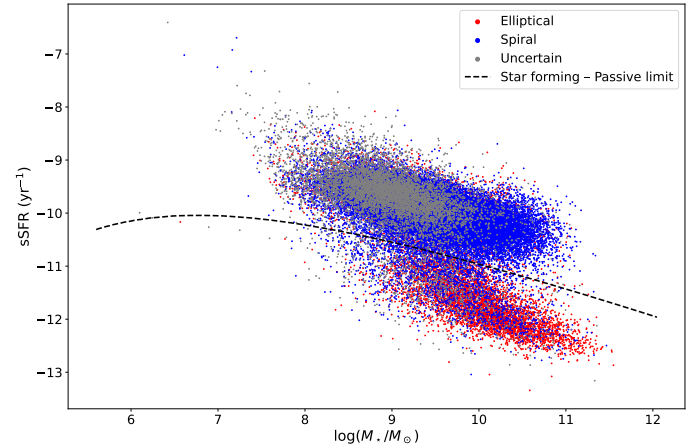


Fig. 3: sSFR vs mass, star-forming/passive limit and Zoo morphologies for the SDSS control sample, excluding quenched galaxies.

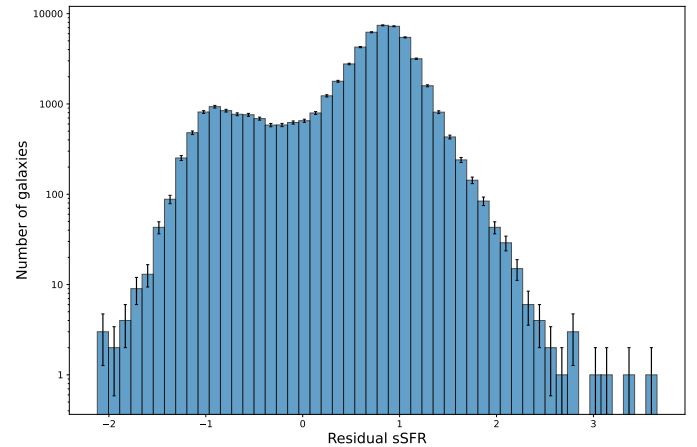


Fig. 4

Control_{4B} sample, 2.5×10^{-1} versus Control_{4C} and 1.8×10^{-2} versus RG₄).

Restrictions of the comparison to BGGs and satellites only are also shown in table 1. The p-values found from two-sided Fisher exact test are 0.58 against Control_{4B}, 0.59 against Control_{4C} and 0.37 against RG₄. There is thus no significant difference in the fraction of star-forming BGGs nor satellites between CG₄ and our control samples either.

in which case it is removed from the non-AGN sample. All remaining galaxies—including those with missing line ratios—are retained as non-AGN. The non-AGN selection process is shown on figure 1.

We identify the galaxies having a `specsfr_tot_p50` of -9999 in the SDSS as quenched (for graphic representation purposes, we give it here an arbitrary low value of $10^{-14.0} \text{ Gyr}^{-1}$). We then consider the other ones and sort them between the star-forming and passive populations using a two-component Gaussian Mixture Model (GMM) in the $\log M_*$ - $\log \text{sSFR}$ plane. Its parameters determined by minimizing the Kullback-Leibler divergence between the empirical and model densities Kullback & Leibler (1951); McLachlan & Peel (2000). For any trial parameter vector θ , we reconstruct the mixture means $\{\mu_i\}$, covariances $\{\Sigma_i\}$, and weights $\{w_i\}$ via a Cholesky-like decomposition and logistic mapping, optionally constraining the second component's mean to isolate the passive population Dempster et al. (1977). The KL divergence is estimated by constructing a 2D histogram of the data and summing $p_{\text{data}} \ln(p_{\text{data}}/p_{\text{GMM}})$, with a small regularization ϵ to avoid singularities Tojeiro et al. (2009); Bisigello et al. (2018). We optimize θ via L-BFGS-B (with fallback to Nelder-Mead) over multiple guided and random initializations, selecting the solution with lowest divergence Pedregosa et al. (2011); Bishop (2006). Initial means are seeded by a median split in $\log \text{sSFR}$ to improve convergence and robustness against local minima Rousseeuw (1987). In the final fit, one component naturally aligns with the high-sSFR “star-forming sequence,” while the other captures the low-sSFR “passive” cloud, providing a data-driven division that agrees with previous multi-Gaussian decompositions of the $\text{SFR}-M_*$ plane Wuyts et al. (2011); Hahn et al. (2019). The decision boundaries, defined as the loci where the posterior probabilities of adjacent components are equal, provide an objective criterion for delineating star-forming from passive galaxies.

Add relevant GM parameters found by the algorithm.

The sSFR status for each sample is shown in table 1. There is no significant differences of for Star forming proportion between CG₄ and the control samples (probabilities estimated using Fisher exact test are 6.5×10^{-1} versus

Sample		Quenched	Passive	Starforming
CG ₄		18 (7.3 %)	146 (58.9 %)	84 (33.9 %)
	<i>BGG</i>	6 (9.7 %)	45 (72.6 %)	11 (17.7 %)
	<i>Satellites</i>	12 (6.5 %)	101 (54.3 %)	73 (39.2 %)
Control _{4B}		181 (6.5 %)	1595 (57.0 %)	1020 (36.5 %)
	<i>BGG</i>	58 (8.3 %)	534 (76.4 %)	107 (15.3 %)
	<i>Satellites</i>	123 (5.9 %)	1061 (50.6 %)	913 (43.5 %)
Control _{4C}		191 (6.3 %)	1619 (53.8 %)	1198 (39.8 %)
	<i>BGG</i>	65 (8.6 %)	569 (75.7 %)	118 (15.7 %)
	<i>Satellites</i>	126 (5.6 %)	1050 (46.5 %)	1080 (47.9 %)
RG ₄		5 (2.2 %)	101 (45.1 %)	118 (52.7 %)
	<i>BGG</i>	3 (5.4 %)	38 (67.9 %)	15 (26.8 %)
	<i>Satellites</i>	2 (1.2 %)	63 (37.5 %)	103 (61.3 %)
<i>SDSS selection</i>		634 (1.2 %)	7705 (14.7 %)	44192 (84.1 %)

Table 1: Number of galaxies in each sSFR status for each sample.

2.3. Morphology

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions [Lintott et al. \(2011\)](#). Those fractions are provided in the `zooSpecz` table of the SDSS database through the `p_el_debiased` and `p_cs_debiased` columns. We attribute to galaxies the morphology “elliptical” if the `p_el_debiased` value is greater than 0.5, and “spiral” if the `p_cs_debiased` value is greater than 0.5. Other cases are considered as “uncertain”. The morphologies found for each sample are shown in table 2. Barnard two-sided exact tests were performed to compare the fraction of each morphology between CG₄ and each control sample. The p-values found are for ellipticals and for spirals versus Control_{4B} sample, and versus Control_{4C} and and versus RG₄. Morphologies of CG₄, which we identified as cores of richer groups in [Tricottet et al. \(2025\)](#), have morphologies significantly different from ordinary groups of four members. Figure 5 displays the sSFR against masses for galaxies from the SDSS selection and for CG₄s galaxies.

2.4. Morphology vs sSFR

Starforming CG₄ are composed of their galaxies classified as spiral (% once galaxies having uncertain morphology removed) and (% once galaxies having uncertain morphology removed) elliptical. These figures are (%) spiral and (%) elliptical galaxies for Control_{4B}, (%) spiral and (%) elliptical galaxies for Control_{4C} and (%) spiral and (%) elliptical galaxies for RG₄. The probability that those fractions come from the same distribution is estimated from the as against Control_{4B}, against Control_{4C} and against RG₄. There is thus neither significant excess nor lack of starforming elliptical galaxies in CG₄ compared to all our control samples.

If we restrict the comparison to BGGs only, we find Elliptical BGGs (% of the BGGs) and Spiral BGGs (%) in CG₄. This is to be compared to (%) Elliptical and (%) Spiral galaxies in Control_{4B}, (%) Elliptical and (%) Spiral in Control_{4C}, and (%) Elliptical and (%) Quenched in RG₄. The p-values found from are against Control_{4B}, against Control_{4C} and against RG₄. There is thus no significant difference in the fraction of star-forming BGGs between CG₄ and our control samples.

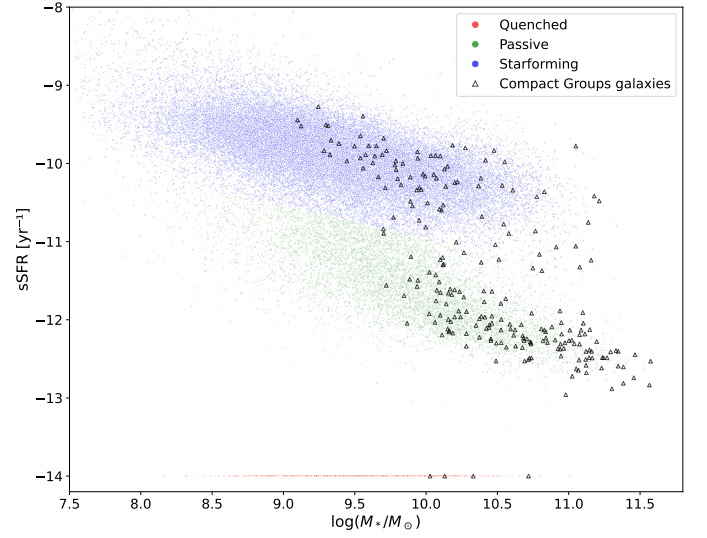


Fig. 5: sSFR and morphologies of galaxies in CG₄ and our SDSS selection. Quenched galaxies are represented at an arbitrary low value of $10^{-14.0} \text{ Gyr}^{-1}$.

3. sSFR

3.1. Main sequence

We model the star-forming main sequence as a power law in stellar mass, to various orders (1 to 4) - cf figure 9. We then choose to use order 2 to compute the residuals of each star-forming galaxy to the main sequence, and plot their histograms in figure 10.

The median main-sequence residual of CG₄ galaxies is significantly different

from that of Control_{4B}, with a median offset of $\Delta = 0.057$ ($0.028 \leq \Delta \leq 0.098$, bootstrap), corresponding to a p-value of $p = 0.048$.

The median main-sequence residual of CG₄ galaxies is significantly different

from that of Control_{4C}, with a median offset of $\Delta = 0.082$ ($0.062 \leq \Delta \leq 0.121$, bootstrap), corresponding to a p-value of $p = 0.009$.

The median main-sequence residual of CG₄ galaxies is not significantly different

Sample	Elliptical	Spiral	Uncertain
CG ₄	124 (%)	86 (%)	38 (%)
Control _{4B}	1107 (%)	1268 (%)	421 (%)
Control _{4C}	1215 (%)	1251 (%)	542 (%)
RG ₄	60 (%)	129 (%)	35 (%)
<i>SDSS selection</i>	<i>10091 (%)</i>	<i>34715 (%)</i>	<i>7725 (%)</i>

Table 2: Number of galaxies of each morphology for each sample.

Sample	$\langle \log sSFR \rangle$		$\langle M_r \rangle$		$\langle \log(M_\star/M_\odot) \rangle$	
	Passive	Star-forming	Passive	Star-forming	Passive	Star-forming
CG4	-12.15	-10.15	-21.04	-20.18	10.65	9.99
Control4B	-12.18	-10.28	-21.68	-20.71	10.87	10.27
Control4C	-12	-10.14	-20.98	-19.71	10.55	9.75
RG4	-11.92	-10.1	-21.31	-20.12	10.76	9.94

Table 3: Median values of galaxy properties for passive and star-forming populations in each sample. Error bars are computed via bootstrapping (10 000 iterations).

Sample	sSFR (yr ⁻¹)		M_r		$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	-11.99 ± 0.09	-10.75 ± 0.29	-20.84 ± 0.11	-20.66 ± 0.32	10.46 ± 0.06	10.28 ± 0.12
Control4B	-11.92 ± 0.03	-10.7 ± 0.06	-21.32 ± 0.03	-21.11 ± 0.07	10.69 ± 0.02	10.54 ± 0.04
Control4C	-11.53 ± 0.03	-10.5 ± 0.06	-20.27 ± 0.05	-20.56 ± 0.08	10.18 ± 0.02	10.24 ± 0.06
RG4	-11.36 ± 0.19	-10.39 ± 0.16	-20.69 ± 0.14	-20.51 ± 0.28	10.3 ± 0.07	10.18 ± 0.13

Table 4: Median values of various quantities for galaxies according to the fertility of their BGG (BGG included). Error bars are computed via bootstrapping (10 000 iterations).

180 from that of RG4, with a median offset of $\Delta = 0.046$
 ($-0.035 \leq \Delta \leq 0.097$, bootstrap), corresponding to a p -value
 of $p = 0.531$.

3.2. Dominance

We split groups into two categories: the dominated, for which the BGG has a luminosity fraction greater than the median of RG₄.

Sample	sSFR (yr^{-1})		M_r	Star-forming BGG	$\log(M_\star/M_\odot)$	
	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG	Passive BGG	Star-forming BGG
CG4	-11.56 ± 0.16	-10.84 ± 0.48	-20.41 ± 0.13	-20.28 ± 0.18	10.2 ± 0.05	10.12 ± 0.06
Control4B	-11.46 ± 0.06	-10.68 ± 0.12	-20.98 ± 0.03	-20.77 ± 0.06	10.49 ± 0.02	10.37 ± 0.03
Control4C	-11 ± 0.04	-10.37 ± 0.07	-19.72 ± 0.04	-20.04 ± 0.10	9.9 ± 0.02	9.9 ± 0.07
RG4	-10.64 ± 0.19	-10.31 ± 0.22	-20.15 ± 0.16	-20.24 ± 0.13	10.08 ± 0.08	10.06 ± 0.13

Table 5: Median values of various quantities for galaxies according to the fertility of their BGG (satellites only). Error bars are computed via bootstrapping (10 000 iterations).

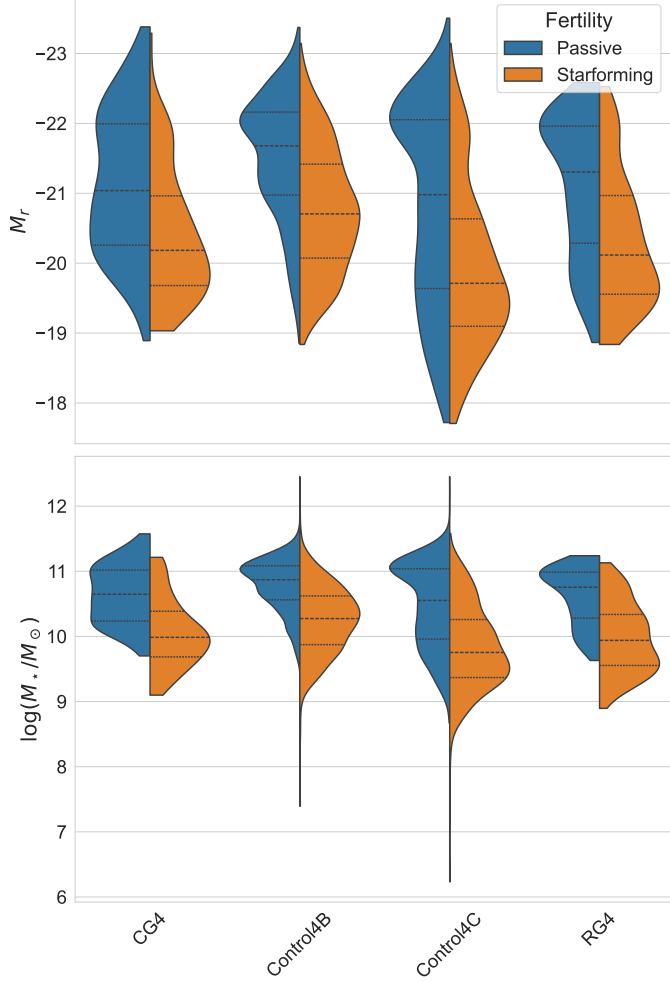


Fig. 6: Distribution various quantities for galaxies according their fertility.

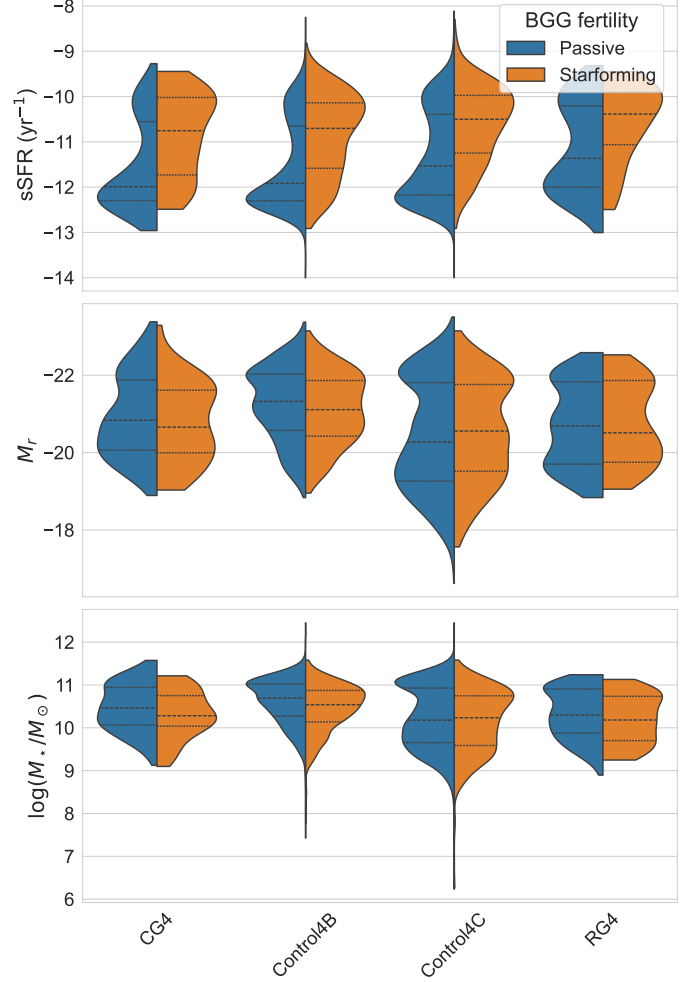


Fig. 7: Distribution various quantities for galaxies according to the fertility of their BGG (BGG included).

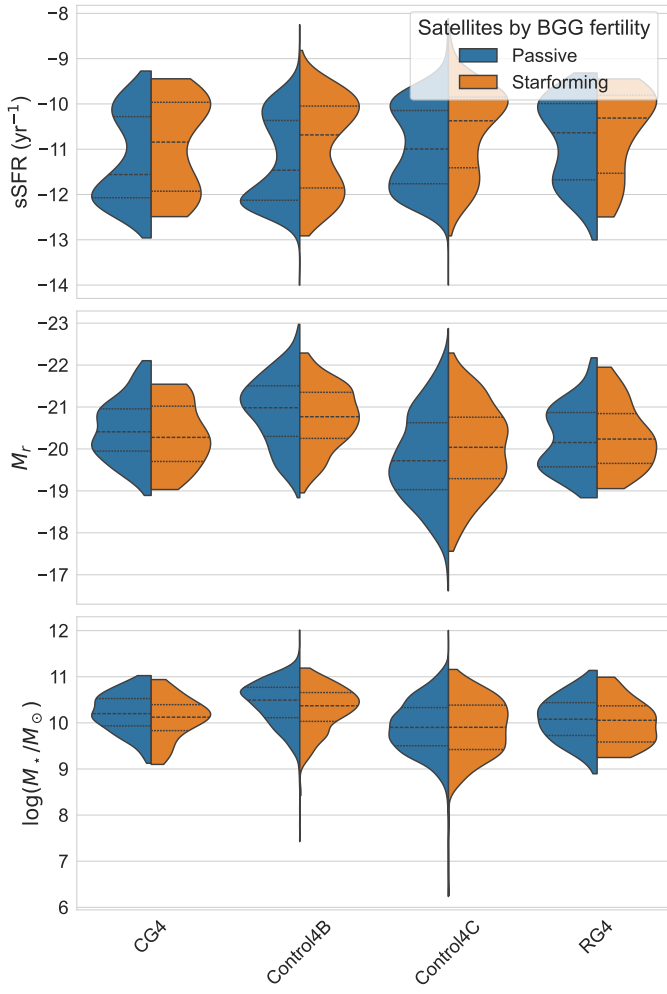


Fig. 8: Distribution various quantities for galaxies according to the fertility of their BGG (BGG excluded).

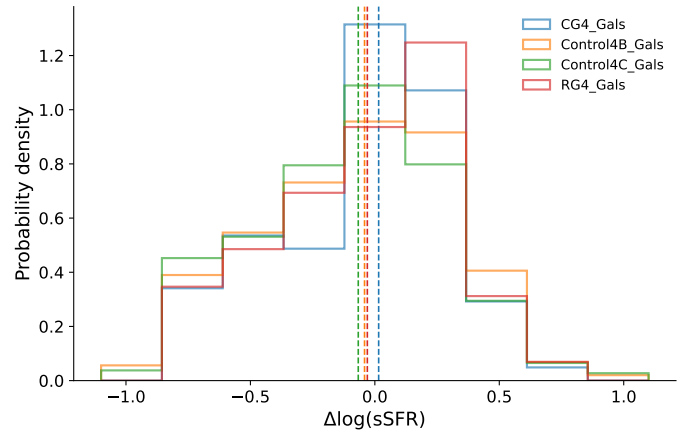


Fig. 10

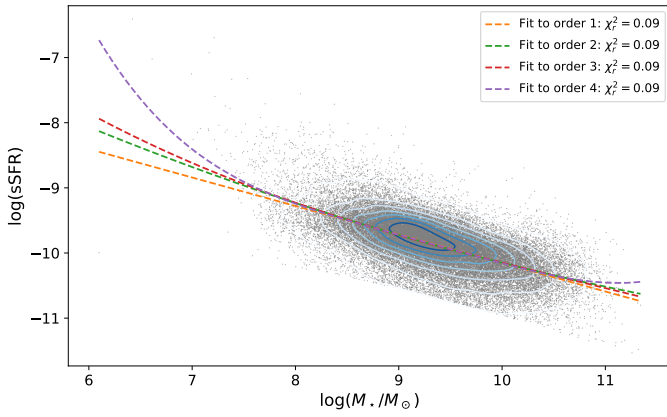


Fig. 9

Interpolation points for sSFR classification

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